

CLAPS: A Multi-LLM Framework for Predicting Social Anxiety via Cognitive-Behavioral Theory of Social Anxiety in VR

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Abstract

Virtual Reality Exposure Therapy (VRET) has become an accessible alternative to in-person exposure therapy for Social Anxiety Disorder (SAD). However, current anxiety assessments in VRET often rely on post-exposure self-reports, increasing the assessment burden for individuals with SAD and limiting objective measurement. Recent studies have proposed prediction models using physiological and behavioral signals; however, these approaches overlook cognitive symptoms, a core component of the cognitive-behavioral theory of social anxiety. In this paper, we introduce *CLAPS* (Cognitive-behavioral LLM-Assisted Prediction of Social anxiety). *CLAPS* operates through two key mechanisms: (1) generating a scene-level cognitive symptom using pre-session self-reports, scene contexts, and symptoms from the preceding scene, and (2) aligning these generated natural-language symptoms with physiological and behavioral time-series data. Experimental results indicate that *CLAPS* outperformed existing modeling approaches, demonstrating the effectiveness of integrating cognitive modeling with multimodal alignment for anxiety prediction in VRET environments. Our *CLAPS* framework code is publicly available: <https://github.com/KDohoon/CLAPS>

CCS Concepts

• Human-centered computing → Virtual reality; • Computing methodologies → Natural language processing; • Applied computing → Life and medical sciences.

Keywords

Social Anxiety Disorder, Large Language Model, Virtual Reality Exposure Therapy, Time-series Data, Theory-based Modeling

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1 Introduction

The global prevalence of anxiety disorders increased by 64.9% between 1990 and 2023 [24]. Among these disorders, Social Anxiety Disorder (SAD) is a mental health disorder characterized by severe anxiety in social situations involving potential evaluation by others. Individuals with SAD tend to withdraw from social interactions to avoid experiencing distressing anxiety symptoms [6, 8]. When such avoidance behaviors persist over time, individuals with SAD may find it increasingly difficult to engage in everyday social activities [15, 23]. Therefore, developing approaches to support the early mitigation and management of SAD is important at both individual and societal levels.

Exposure therapy, one of the standard treatments for SAD, reduces avoidance behavior by systematically exposing individuals to anxiety-inducing stimuli. Recently, Virtual Reality Exposure Therapy (VRET) has emerged as a promising alternative to overcome the temporal and spatial limitations of conventional in-person exposure therapy [17]. In VRET settings, post-exposure self-report questionnaires (e.g., PEPQ-R, STAI) are often used to measure the level of anxiety during treatment [12]. However, these self-reports impose cognitive and psychological burdens and are susceptible to retrospective bias in the reporting of experienced anxiety [2, 18, 25]. Consequently, there is a clear need for objective and automated methods to assess anxiety within VRET environments [16, 27].

According to the cognitive-behavioral theory of social anxiety, individuals experiencing anxiety in social situations exhibit a combination of physiological, behavioral, and cognitive symptoms [22]. Precisely measuring anxiety therefore requires a comprehensive consideration of all three components. However, existing prediction models have mainly focused on physiological and behavioral symptoms. For example, studies have predicted the level of anxiety using machine learning models (e.g., XGBoost, LightGBM) and deep learning models (e.g., GRU, LSTM), focusing on physiological symptoms (e.g., heart rate) and behavioral cues (e.g., gaze) [7, 9, 21, 26, 29]. On the other hand, internal cognitive symptoms remain largely underexplored in data modeling, primarily because they are difficult to observe externally or quantify via wearable sensor devices [10, 14].

In this context, the emergence of Large Language Models (LLMs)—capable of understanding complex contexts and inferring conceptual meanings—presents new opportunities to capture and model previously inaccessible cognitive symptoms. LLMs have demonstrated substantial potential in understanding a user’s situational contexts and personal characteristics from textual information and in deriving personalized psychological inferences. For example, Kim et al. [13] proposed Health-LLM, which leverages individual attributes (e.g., emotion, sleep duration) alongside

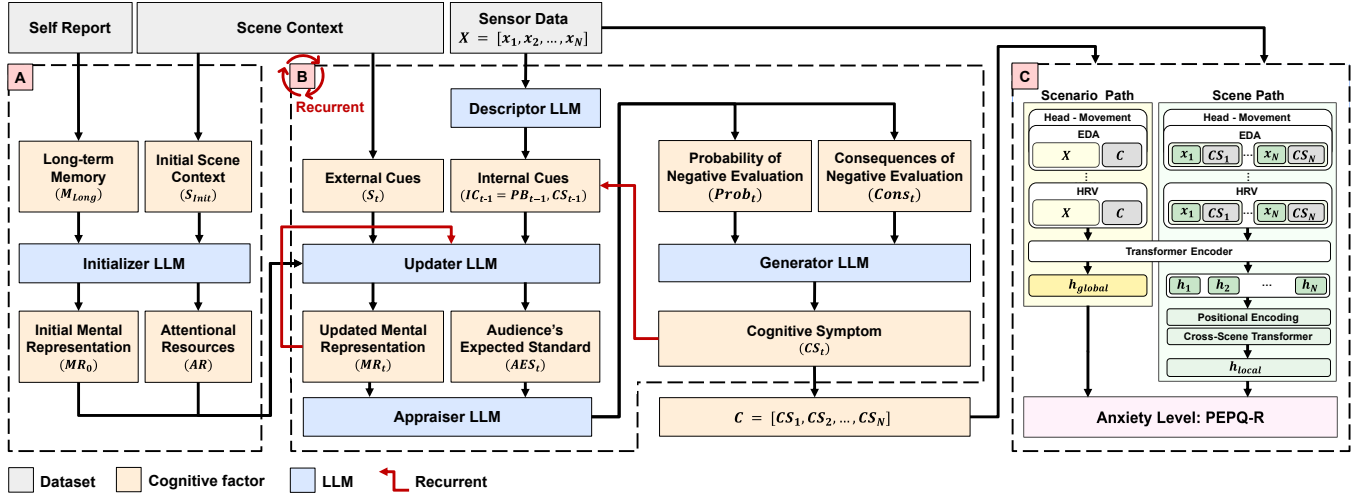


Figure 1: The CLAPS framework. (A) Initialization Module, which generates initial cognitive factors. (B) Cognitive Symptom Generation Module, which updates cognitive factors and generates a scene-level cognitive symptom. (C) Cross-modal Aligned Transformer (CAT) Module, which aligns multimodal symptoms and predicts PEPQ-R scores.

clinical domain knowledge to predict mental health status. Zhang et al. [28] developed NarrativeSense, which captures an individual’s behavioral context from smartphone logs (e.g., app usage) and interprets it with LLMs for mental health assessment.

Importantly, effectively leveraging LLMs to generate cognitive symptoms and integrate them into anxiety prediction requires addressing two key considerations. First, the underlying cognitive processes should be precisely modeled based on psychological theory. Encoding the mechanisms described in the cognitive-behavioral theory of social anxiety into a modular architecture can systematically guide the LLM to yield reliable cognitive symptoms. Second, multimodal alignment across heterogeneous data streams is needed. Time-series data representing physiological and behavioral symptoms captured from sensor signals should be aligned with natural language data representing cognitive symptoms derived by the LLMs within the co-occurring spatiotemporal context. This alignment not only enhances prediction performance but also establishes the interpretability of the model’s outputs.

To address these requirements, we propose *CLAPS* (**C**ognitive-behavioral **LLM**-**A**ssisted **P**rediction of **S**ocial anxiety), a multimodal framework grounded in the cognitive-behavioral theory of social anxiety [22]. *CLAPS* consists of three core modules: (1) the Initialization Module, which establishes an individual’s initial mental representation and preferential allocation of attentional resources at the point of entering a social situation based on pre-session self-reports and initial scene context; (2) the Cognitive Symptom Generation Module, which updates these mental representations and the appraisals of the audience’s expected standard as the situation progresses, and generates cognitive symptoms by inferring the probability and consequences of negative evaluation; and (3) the Cross-modal Aligned Transformer (CAT) Module, which aligns the generated cognitive symptom with co-occurring physiological and behavioral time-series data at the scene level to predict anxiety scores measured by the Post-Event Processing Questionnaire-Revised (PEPQ-R) [19].

We validate our framework using a dataset collected from a VRET scenario designed to induce social anxiety [12]. Using physiological and behavioral time-series data, pre-session self-report information, and scene-level contextual information, *CLAPS* generated cognitive symptoms and predicted scores for nine PEPQ-R items rated on a 7-point Likert scale. *CLAPS* achieved an MAE of 1.268 and an RMSE of 1.558, outperforming mental health prediction models proposed in previous studies. These results suggest that the approach of generating cognitive symptoms through theory-driven process modeling and aligning them with co-occurring physiological and behavioral data is effective for anxiety prediction. Furthermore, given that symptoms of other psychiatric conditions—such as obsessive-compulsive disorder and depression—manifest through similar cognitive processes [3], our theory-based modeling approach holds potential for extension to broader mental health domains.

2 The CLAPS Framework

CLAPS is grounded in the cognitive-behavioral theory of social anxiety [22]. Based on the theory, we decomposed the cognitive process into four stages and operationalized each stage through LLM-driven step-by-step reasoning [11]. This theory-guided design enables *CLAPS* to model how anxiety responses develop through the cognitive processes described by the theory. In this section, we explain how the four stages of the cognitive process, their recurrent dynamics across scenes, and the final anxiety prediction are operationalized across the three modules of *CLAPS*.

2.1 Initialization Module

The Initialization Module consists of a single Initializer LLM, denoted as L_{Init} (Figure 1-(A)). When an individual enters a social situation, L_{Init} generates the initial mental representation (MR_0) and the preferential allocation of attentional resources (AR) based on the individual’s long-term memory (M_{Long}), which includes past social experiences and characteristics related to anxiety, and the initial scene context (S_{init}), including audience information.

$$(MR_0, AR) = L_{Init}(M_{Long}, S_{init}) \quad (1)$$

2.2 Cognitive Symptom Generation Module

The Cognitive Symptom Generation Module is implemented with three distinct LLMs task-engineered to generate a cognitive symptom arising in each scene of a social situation (Figure 1-(B)).

2.2.1 Updater LLM. The Updater LLM (L_{Update}) takes four inputs: the mental representation from the previous scene (MR_{t-1}), the preferential allocation of attentional resources (AR), the external cue in the current scene (S_t), and the internal cue from the previous scene (IC_{t-1}). Based on these inputs, L_{Update} updates the mental representation of the current scene (MR_t) and the appraisal of the audience's expected standard (AES_t).

$$IC_{t-1} = \begin{cases} \emptyset & t = 1 \\ (PB_{t-1}, CS_{t-1}) & t \geq 2 \end{cases} \quad (2)$$

$$(MR_t, AES_t) = \begin{cases} L_{Update}(MR_0, AR, S_t) & t = 1 \\ L_{Update}(MR_{t-1}, AR, S_t, IC_{t-1}) & t \geq 2 \end{cases}$$

where, t denotes the index of the scene order. PB_{t-1} is a natural language summary of time-series changes in physiological and behavioral symptoms of the previous scene. For the first scene ($t = 1$), since the previous scene ($t - 1$) does not exist, the internal cue is defined as $\emptyset$, and the initial mental representation (MR_0) generated by L_{Init} is used as the mental representation of the first scene (MR_1).

2.2.2 Appraiser LLM. The Appraiser LLM ($L_{Appraise}$) generates the probability of negative evaluation from the audience ($Prob_t$) and the consequences of negative evaluation ($Cons_t$) based on the mental representation of the current scene (MR_t) and the appraisal of the audience's expected standard (AES_t).

$$(Prob_t, Cons_t) = L_{Appraise}(MR_t, AES_t) \quad (3)$$

2.2.3 Generator LLM. The Generator LLM ($L_{Generate}$) generates the cognitive symptom of the current scene (CS_t) based on the probability of negative evaluation from the audience ($Prob_t$) and the consequences of negative evaluation ($Cons_t$).

$$CS_t = L_{Generate}(Prob_t, Cons_t) \quad (4)$$

2.2.4 Recurrent Structure. In the recurrent structure of CLAPS, the physiological and behavioral symptoms (x_t) collected as time-series data in each scene (t) are used as the internal cue of the next scene. However, since LLMs process input as token sequences, they struggle to interpret time-series changes [4, 20]. To address this limitation, we use the Descriptor LLM ($L_{Describe}$) to generate PB_t , a natural language description of changes in x_t .

$$PB_t = L_{Describe}(x_t) \quad (5)$$

These generated descriptions PB_t and CS_t are used as the internal cue (IC_t) for L_{Update} in the next scene ($t+1$). This recurrent structure sequentially generates a scene-level cognitive symptom (CS_t) to form the cognitive symptom sequence C across the entire scenario.

2.3 Cross-modal Aligned Transformer Module

The Cross-modal Aligned Transformer (CAT), comprising the Scenario Path and Scene Path, predicts PEPQ-R scores by aligning time-series-based physiological and behavioral symptoms with cognitive symptoms at both scenario and scene levels (Figure 1-(C)). The Scenario Path ($P_{Scenario}$) separately embeds time-series data over

the entire scenario ($X = [x_1, \dots, x_N]$) and the cognitive symptom sequence ($C = [CS_1, \dots, CS_N]$), and generates a global embedding (h_{global}) that reflects the overall pattern of the scenario.

$$h_{global} = P_{Scenario}(Emb_{patch}(X), Emb_{text}(C)) \quad (6)$$

The Scene Path (P_{Scene}) generates the scene embedding (h_t) by aligning the time-series data (x_t) and the cognitive symptom (CS_t) from the same scene.

$$h_t = P_{Scene}(Emb_{patch}(x_t), Emb_{text}(CS_t)), \quad t = 1, \dots, N \quad (7)$$

where, $Emb_{patch}(\cdot)$ embeds patched time-series data and $Emb_{text}(\cdot)$ embeds cognitive symptom text.

Then, positional encoding (PE) and the Cross-Scene Transformer (CST) encode the scene-level embedding sequence $[h_1, \dots, h_N]$ into the local embedding h_{local} , capturing cross-scene relationships.

$$h_{local} = CST(PE([h_1, h_2, \dots, h_N])) \quad (8)$$

Finally, CAT combines the global embedding (h_{global}) and the local embedding (h_{local}) to predict the PEPQ-R scores.

3 Experiments

3.1 Dataset

To validate the performance of the CLAPS, we used a publicly available dataset [12]. This dataset includes two scenarios (self-introduction and job interview) and contains 30 time-series features encompassing physiological signals, head movement signals, and eye tracking signals collected from 32 participants. The dataset also includes participants' pre-session self-reports on Self-Presentation Motivation (SPM) and Self-Presentation Expectancy (SPE), as well as post-scenario responses to PEPQ-R.

To operationalize long-term memory in the Initialization Module, we used SPM, SPE, and the derived discrepancy scores $Diff_{(SPM-SPE)}$, which is defined as $SPM \times (1 - SPE)$. We also constructed a text-based social context description consisting of scene context, audience information, task, and the audience's nonverbal responses to represent the external cue in each scene.

3.2 Model Setup

We used GPT-4o mini for all LLM-based components in CLAPS. In the CAT Module, we used KURE-v1 fine-tuned from BGE-M3 [5] to encode the generated cognitive symptom text into vector representations. The patch size and stride for time-series data were set to 32 and 16, respectively. For training, we used the AdamW optimizer with a batch size of 16, a learning rate of 0.0001, and 100 epochs.

3.3 Baseline Models

We selected six competitive baseline models. First, we chose Random Forest, XGBoost, and LightGBM, which are widely established in the mental health domain [7, 21, 26]. These models were trained using statistical features (e.g., mean) from time-series data. Second, we selected SAFER [9] and DepGuard [29] as deep learning-based mental health prediction baselines that learn temporal patterns from time-series data. These models were included to evaluate performance improvements when explicitly considering temporal dynamics and cross-modality relationships. Lastly, we selected Health-LLM [13] as an LLM-based mental health prediction model that uses individual characteristics and the mental health domain knowledge in the prediction process.

Table 1: Performance comparison of *CLAPS* and baseline. All scores are averaged over 8-fold cross-validation. *CLAPS* w/o Cognitive Symptom removes the Initialization and Cognitive Symptom Generation module, and *CLAPS* w/o Scene Path removes scene-level alignment across symptoms. Statistical significance of the gap between *CLAPS* and each model is assessed by a two-sided paired Wilcoxon signed-rank test across 64 test samples (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$). **Bold: best performance, underlined: second-best performance.****

Model	MAE	RMSE
Random Forest	1.745 $\pm$ (0.278)***	2.300 $\pm$ (0.270)***
XGBoost	1.703 $\pm$ (0.201)***	2.235 $\pm$ (0.221)***
LightGBM	1.731 $\pm$ (0.235)***	2.228 $\pm$ (0.242)***
SAFER [9]	1.459 $\pm$ (0.271)**	1.769 $\pm$ (0.274)**
DepGuard [29]	1.384 $\pm$ (0.218)	1.728 $\pm$ (0.227)**
Health-LLM [13]	1.564 $\pm$ (0.327)***	1.995 $\pm$ (0.336)***
<i>CLAPS</i> w/o Cognitive Symptom	1.369 $\pm$ (0.223)	1.664 $\pm$ (0.271)
<i>CLAPS</i> w/o Scene Path	1.344 $\pm$ (0.172)**	1.619 $\pm$ (0.201)**
<i>CLAPS</i> (ours)	1.268 $\pm$ (0.180)	1.558 $\pm$ (0.183)

3.4 Experimental Results

As shown in Table 1, *CLAPS* achieved the best performance among all models, yielding an MAE of 1.268 and an RMSE of 1.558. This result supports the effectiveness of our approach compared to conventional machine learning, deep learning, and existing LLM-centric modeling techniques. *CLAPS* also showed performance improvements of 7.38% in MAE and 6.37% in RMSE over *CLAPS* without Cognitive Symptoms, and 5.65% in MAE and 3.77% in RMSE over *CLAPS* without the Scene Path. These results suggest that the generated cognitive symptoms provide additional information for predicting social anxiety, and show the importance of aligning physiological, behavioral, and cognitive symptoms at the granular, scene level.

4 Discussion

4.1 Performance Implications

As shown in Section 3.4, *CLAPS* achieved significantly higher performance than the baseline models. The ablation study further showed that removing the cognitive symptom component resulted in higher prediction errors, although the differences were not statistically significant. The non-significant difference may reflect the predictive capacity of the CAT module alone. However, the consistent improvement observed with dynamically generated cognitive symptoms supports their contribution to anxiety prediction. Consequently, our findings highlight the value of capturing cognitive symptoms for predicting anxiety and the potential of LLMs to infer these internal states from contextual cues.

Furthermore, *CLAPS* achieved enhanced performance through the integration of both the Scenario and Scene Paths. This result indicates that aligning physiological and behavioral symptoms with the cognitive symptoms co-occurring within the exact same scene boosts predictive power. Since social anxiety is a psychological phenomenon that accumulates over time [22], robust anxiety prediction frameworks need to account for both the macro-level narrative of a scenario and the sequential, scene-level temporal dependencies where cognitive interpretations unfold.

4.2 Limitations and Future Work

Although *CLAPS* has demonstrated effectiveness in predicting social anxiety, we acknowledge three primary limitations that warrant further investigation.

First, to capture an individual’s long-term memory and generate scene-level cognitive symptoms, *CLAPS* currently depends on self-reports and structured social context descriptions. Due to such data requirements, the framework may have limited applicability. Our future work will explore the integration of Vision-Language Models (VLMs) to automatically synthesize social context descriptions by analyzing audience positioning, gaze dynamics, utterances, and behavioral reactions directly within VRET environments.

Second, because the architecture of *CLAPS* is explicitly tailored to the cognitive-behavioral theory of social anxiety, it may not be transferable to other psychological domains without modification. For example, in depressive symptoms, the dominant cognitive process involves self-blame—wherein individuals attribute negative events entirely to themselves [1]. Adapting *CLAPS* to encompass other conditions would therefore necessitate distinct theoretical foundations and subsequent structural modifications. Nevertheless, our empirical results validate that core theoretical perspectives can be successfully operationalized and inferred using LLMs. Future research will focus on developing a more modular framework capable of flexibly incorporating diverse cognitive-behavioral paradigms, which we intend to evaluate using large-scale, heterogeneous datasets.

Third, the cognitive symptoms generated by *CLAPS* may not necessarily be clinically valid or accurately reflect users’ actual cognitive experiences. Therefore, future work should evaluate the theoretical and clinical validity of the generated cognitive symptoms through expert assessment, and examine their correspondence with users’ actual cognitive experiences through user studies.

5 Conclusion

In this paper, we proposed *CLAPS*, a multi-LLM framework that predicts anxiety levels by generating cognitive symptoms grounded in the cognitive-behavioral theory of social anxiety and aligning them with co-occurring physiological and behavioral data. We showed that *CLAPS* outperformed all baseline models, suggesting that integrating an individual’s cognitive interpretation of social situations with physiological and behavioral symptoms is important for anxiety prediction. Furthermore, our findings suggest that aligning the temporal progression across symptoms is effective for anxiety prediction. This work provides valuable insights into the design of automated, theory-driven anxiety prediction systems within VRET environments.

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